Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Cubic functions**

Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. Given that $P(1) = 10$, $P(2) = 27$, $P(3) = 58$, and $P(4) = 115$, determine the value of a.

Let the polynomial be $P(x) = ax^3 + bx^2 + cx + d$. We are given the following values:

$P(1) = a(1)^3 + b(1)^2 + c(1) + d = a + b + c + d = 10$ (1) $P(2) = a(2)^3 + b(2)^2 + c(2) + d = 8a + 4b + 2c + d = 27$ (2)

$P(3) = a(3)^3 + b(3)^2 + c(3) + d = 27a + 9b + 3c + d = 58$ (3) $P(4) = a(4)^3 + b(4)^2 + c(4) + d = 64a + 16b + 4c + d = 115$ (4) Consider the differences between consecutive terms: $P(2) - P(1) = (8a + 4b + 2c + d) - (a + b + c + d) = 7a + 3b + c = 27 - 10 = 17$ (5).

2 **[Foundational]** • **Target: Power functions**

Prove by direct deductive reasoning that the difference between the squares of any two odd integers is always divisible by 8.

1. Let m and n be any two odd integers. Then they can be expressed algebraically as $m = 2k + 1$ and $n = 2j + 1$ for some integers $k, j \in \mathbb{Z}$. 2. Form the difference of their squares:

$m^2 - n^2 = (2k + 1)^2 - (2j + 1)^2 = (4k^2 + 4k + 1) - (4j^2 + 4j + 1) = 4k^2 + 4k - 4j^2 - 4j = 4[k(k + 1) - j(j + 1)]$. 3. Analyze the products $k(k + 1)$ and $j(j + 1)$: For any integer k, exactly one of k or $k + 1$ is even. Therefore, the product of any two consecutive integers is always divisible by 2.

3 **[Foundational]** • **Target: Direct variation**

The sum of the first n terms of an arithmetic series is given by $S_n = 4n^2 - 3n$. (a) Find the first term u_1 and the second term u_2. (b) Find the common difference d. (c) Find an explicit expression for the nth term u_n, and calculate u_{25}.

(a) $u_1 = S_1 = 4(1)^2 - 3(1) = 1$. $S_2 = 4(2)^2 - 3(2) = 16 - 6 = 10$. $u_2 = S_2 - S_1 = 10 - 1 = 9$. (b) Common difference: $d = u_2 - u_1 = 9 - 1 = 8$. (c) General term:

$u_n = u_1 + (n - 1)d = 1 + (n - 1)(8) = 8n - 7$. Alternatively:

$u_n = S_n - S_{n-1} = (4n^2 - 3n) - [4(n - 1)^2 - 3(n - 1)] = 8n - 7$. For $n = 25$: $u_{25} = 8(25) - 7 = 200 - 7 = 193$.

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4 [Foundational] • Target: Hyperbolas and inverse variation

A particle's position on the x-axis at time t (in seconds) is given by the function $x(t) = t^3 - 6t^2 + 9t + 1$. The particle starts at $t = 0$. Determine the time intervals during which the particle is moving in the negative direction and calculate the total distance traveled by the particle during the first 4 seconds.

First, find the velocity function by differentiating the position function with respect to time: $v(t) = x'(t) = 3t^2 - 12t + 9$. To find when the particle is moving in the negative direction, we set $v(t) < 0$. Factor the velocity function: $v(t) = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3)$. Thus, $v(t) < 0$ when $1 < t < 3$. So, the particle moves in the negative direction during the interval $(1, 3)$ seconds. Next, calculate the total distance traveled during the first 4 seconds. This requires finding the positions at criti.

5 [Foundational] • Target: Transformations of hyperbolas

Consider a geometric progression where the sum of the first n terms is given by $S_n = \frac{3(2^n - 1)}{2}$. A second sequence, denoted by a_n, is defined such that $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$. Find the value of $a_5 - a_3$.

The sum of the first n terms of the geometric progression is $S_n = \frac{3(2^n - 1)}{2}$. The n-th term of the sequence, a_n, can be found using the relation $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$. First, let's find a_1: $a_1 = S_1 = \frac{3(2^1 - 1)}{2} = \frac{3(1)}{2} = \frac{3}{2}$. Now, let's find a general formula for a_n for $n \geq 2$: $S_{n-1} = \frac{3(2^{n-1} - 1)}{2}$. $a_n = S_n - S_{n-1} = \frac{3(2^n - 1)}{2} - \frac{3(2^{n-1} - 1)}{2}$.

6 [Foundational] • Target: Graphing square root functions

In triangle ABC, side $a = 7$ cm, side $b = 9$ cm, and angle $A = 42^\circ$. (a) Show that two distinct triangles can be formed with these given measurements. (b) Calculate both possible values for angle B and the corresponding values of side c.

(a) Height of triangle from vertex C: $h = b\sin A = 9\sin 42^\circ \approx 9(0.6691) = 6.022$ cm. Since $h < a < b$ (i.e. $6.022 < 7 < 9$), the side a is long enough to reach the base line on either side of the altitude. Therefore, exactly two non-congruent triangles exist. ■ (b) Sine rule: $\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{9\sin 42^\circ}{7} \approx \frac{6.0221}{7} \approx 0.8603$. First triangle (acute): $B_1 = \arcsin(0.8603) \approx 59.35^\circ \approx 59.4^\circ$..

7 [Foundational] • Target: Cubic functions

Three mobile antennas are located at coordinates $A(0, 0)$, $B(6, 0)$, and $C(2, 4)$. (a) Find the equation of the perpendicular bisector of segment AB. (b) Find the equation of the perpendicular bisector of segment AC. (c) Find the coordinates of the Voronoi vertex (circumcentre) equidistant from all three antennas, and calculate the radius of the circumcircle.

(a) For segment AB: $A(0, 0)$ and $B(6, 0)$ lie on the x-axis. Midpoint $M_{AB} = (3, 0)$. The perpendicular bisector is the vertical line $x = 3$. (b) For segment AC: $A(0, 0)$ and $C(2, 4)$. Midpoint $M_{AC} = \left(\frac{0+2}{2}, \frac{0+4}{2}\right) = (1, 2)$. Gradient $m_{AC} = \frac{4-0}{2-0} = 2$. Perpendicular gradient $m_\perp = -\frac{1}{2}$. Equation: $y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2} \Leftrightarrow x + 2y = 5$. (c) The Voronoi vertex is the intersectio.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Power functions**

Six wines were evaluated by two independent sommeliers, X and Y, giving the following ratings: Wine: A, B, C, D, E, F Judge X: 88, 72, 95, 60, 81, 75 Judge Y: 84, 78, 91, 65, 85, 70 (a) Construct a table of ranks for both judges. (b) Calculate Spearman's rank correlation coefficient r_s. (c) State what this coefficient indicates about the judges' agreement.

(a) Ranking in descending order (highest rating = rank 1): Judge X: 95 (1), 88 (2), 81 (3), 75 (4), 72 (5), 60 (6). Judge Y: 91 (1), 85 (2), 84 (3), 78 (4), 70 (5), 65 (6). Table of paired ranks and differences $d_i = R_X - R_Y$: Wine A: $R_X = 2, R_Y = 3 \Rightarrow d_1 = -1, d_1^2 = 1$ Wine B: $R_X = 5, R_Y = 4 \Rightarrow d_2 = 1, d_2^2 = 1$ Wine C: $R_X = 1, R_Y = 1 \Rightarrow d_3 = 0, d_3^2 = 0$ Wine D: $R_X = 6, R_Y = 6 \Rightarrow d_4 = 0, d_4^2 = 0$ Wine E: $R_X = 3, R_Y = 2 \Rightarrow d_5 = 1, d_5^2 = 1$.

9 **[Intermediate]** • **Target: Direct variation**

A small business borrows USD 40,000 to renovate its premises. The loan carries an annual nominal interest rate of 7.2% compounded monthly, with equal monthly repayments made at the end of each month over 6 years. (a) Calculate the monthly repayment amount. (b) Calculate the total amount repaid and the total interest charged on the loan. (c) After making payments for 3 full years, determine the outstanding loan balance.

(a) Total periods $N = 6 \times 12 = 72$. Monthly interest rate $r = \frac{7.2\%}{12} = 0.6\% = 0.006$. Using the present value of an annuity formula: $PV = PMT\left[\frac{1-(1+r)^{-N}}{r}\right] \Rightarrow 40,000 = PMT\left[\frac{1-(1.006)^{-72}}{0.006}\right]$. $(1.006)^{-72} \approx 0.65028 \Rightarrow 1 - 0.65028 = 0.34972$. Fraction: $\frac{0.34972}{0.006} \approx 58.2866$. $PMT = \frac{40,000}{58.2866} \approx USD686.26$. (b) Total amount repaid: $72 \times 686.264 = USD49,411.01$. Total intere.

10 **[Intermediate]** • **Target: Hyperbolas and inverse variation**

A particle moves along a straight line such that its velocity in m/s at time t seconds is given by $v(t) = 6t - 2t^2$ for $0 \le t \le 4$. (a) Find the acceleration of the particle at $t = 2$ s. (b) Determine the net displacement of the particle between $t = 0$ and $t = 4$. (c) Calculate the total distance travelled by the particle over the interval $[0, 4]$.

(a) Acceleration: $a(t) = v'(t) = 6 - 4t$. At $t = 2$: $a(2) = 6 - 4(2) = -2$ m/s^2. (b) Net displacement is the integral of velocity: $\Delta s = \int_0^4 (6t - 2t^2)\,dt = \left[3t^2 - \frac{2}{3}t^3\right]_0^4 = (3(16) - \frac{2}{3}(64)) - 0 = 48 - \frac{128}{3} = \frac{16}{3} = 5\frac{1}{3}$ m. (c) Velocity vanishes when $6t - 2t^2 = 0 \Rightarrow 2t(3 - t) = 0 \Rightarrow t = 0$ and $t = 3$. For $0 \le t \le 3$, $v(t) \ge 0$.

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11 **[Intermediate]** • **Target: Transformations of hyperbolas**

A curve has equation $f(x) = \frac{2x^2}{x^2 + k}$ for some positive constant $k > 0$. (a) Write down the horizontal asymptote of $y = f(x)$. (b) Find $f'(x)$ in terms of k. (c) Given that the gradient of the tangent to the curve at $x = 1$ is equal to $\frac{1}{2}$, find all exact possible values of k.

(a) $\lim_{x \to \pm\infty} \frac{2x^2}{x^2 + k} = \lim_{x \to \pm\infty} \frac{2}{1 + k/x^2} = 2$. Horizontal asymptote: $y = 2$. (b) Apply the quotient rule with $u = 2x^2$ and $v = x^2 + k$: $u' = 4x$, $v' = 2x$. $f'(x) = \frac{4x(x^2 + k) - 2x^2(2x)}{(x^2 + k)^2} = \frac{4x^3 + 4kx - 4x^3}{(x^2 + k)^2} = \frac{4kx}{(x^2 + k)^2}$. (c) At $x = 1$, $f'(1) = \frac{1}{2}$: $\frac{4k(1)}{(1^2 + k)^2} = \frac{1}{2} \Rightarrow \frac{4k}{(k+1)^2} = \frac{1}{2} \Rightarrow 8k = (k+1)^2$. Expand.

12 **[Intermediate]** • **Target: Graphing square root functions**

A closed cylindrical metal container with circular base of radius r cm and height h cm has a fixed volume of 250π cm^3. (a) Express h in terms of r. (b) Show that the total surface area is $A(r) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Find the radius r that minimizes the surface area, and show that for this optimal cylinder, the height equals the diameter ($h = 2r$).

(a) Volume of cylinder: $V = \pi r^2 h = 250\pi \Rightarrow h = \frac{250}{r^2}$. (b) Total surface area includes 2 circular ends plus curved surface: $A = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r\left(\frac{250}{r^2}\right) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Differentiate with respect to r: $A'(r) = 4\pi r - \frac{500\pi}{r^2}$. Set $A'(r) = 0$: $4\pi r = \frac{500\pi}{r^2} \Rightarrow 4r^3 = 500 \Rightarrow r^3 = 125 \Rightarrow r = 5$ cm. Check second derivative.

13 **[Intermediate]** • **Target: Cubic functions**

A continuous random variable X is normally distributed with mean μ and standard deviation σ. Given that $P(X < 45) = 0.12$ and $P(X > 85) = 0.08$, find the value of μ and σ correct to 3 significant figures.

1. Standardize using $Z = \frac{X - \mu}{\sigma}$: For the lower tail: $P(X < 45) = 0.12 \Rightarrow P(Z < z_1) = 0.12$. Using GDC `invNorm(0.12, 0, 1)`: $z_1 = -1.17499$. For the upper tail: $P(X > 85) = 0.08 \Rightarrow P(X < 85) = 0.92 \Rightarrow P(Z < z_2) = 0.92$. Using GDC `invNorm(0.92, 0, 1)`: $z_2 = 1.40507$. 2. Form the system of equations: $\frac{45 - \mu}{\sigma} = -1.1750 \Rightarrow \mu - 1.1750\sigma = 45$. $\frac{85 - \mu}{\sigma} = 1.4051 \Rightarrow \mu + 1.4051\sigma = 85$. 3. Subtract the first.

14 **[Intermediate]** • **Target: Power functions**

Consider the function $f(x) = x^2 e^{-x}$. (a) Find $f'(x)$ and express it in factored form. (b) Hence, find the coordinates of all stationary points of f. (c) Use the second derivative test to classify each stationary point as a local maximum or local minimum.

(a) Apply the product rule with $u = x^2$ and $v = e^{-x}$: $u' = 2x$, $v' = -e^{-x}$.
$f'(x) = 2xe^{-x} + x^2(-e^{-x}) = (2x - x^2)e^{-x} = x(2 - x)e^{-x}$. (b) Set $f'(x) = 0$. Since $e^{-x} > 0$ for all x:
$x(2 - x) = 0 \Rightarrow x = 0 \quad \text{or} \quad x = 2$. Coordinates: At $x = 0$: $f(0) = 0^2 e^0 = 0 \Rightarrow (0, 0)$. At $x = 2$:
$f(2) = 2^2 e^{-2} = 4e^{-2} = \frac{4}{e^2} \approx 0.5413 \Rightarrow \left(2, \frac{4}{e^2}\right)$. (c) Compute $f''(x)$ by differentiating $(2x - x^2)e^{-x}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Direct variation** • **Cross-Lesson: 1.01 Number & Algebra**

Find the exact area of the finite region enclosed between the parabola $y = 3x - x^2$ and the line $y = x - 3$.

1. Find intersection points: $3x - x^2 = x - 3 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0 \Rightarrow x = -1 \quad \text{and} \quad x = 3.$ 2. On the interval $[-1, 3]$, the parabola is concave down and lies above the line: At $x = 0$: $y_{\text{parabola}} = 0 > -3 = y_{\text{line}}$. 3. Set up the definite integral: $A = \int_{-1}^{3} [(3x - x^2) - (x - 3)]\,dx = \int_{-1}^{3} (3 + 2x - x^2)\,dx.$ 4. Integrate.

16 **[Challenging]** • **Target: Hyperbolas and inverse variation** • **Cross-Lesson: 3.01 Geometry & Trigonometry**

In triangle PQR, angle PQR = 90 degrees. S is a point on PR such that QS is perpendicular to PR. The length of PQ is 15 cm and the length of QR is 20 cm. Let the length of PS be x cm and the length of SR be y cm. Find the value of x.

First, we find the length of the hypotenuse PR using the Pythagorean theorem in triangle PQR: $PR^2 = PQ^2 + QR^2$. $PR^2 = 15^2 + 20^2 = 225 + 400 = 625$. So, $PR = \sqrt{625} = 25$ cm. Triangle PQR is similar to triangle PSQ (by AA similarity, as both share angle P and have a right angle). Therefore, the ratio of corresponding sides is equal: $\frac{PS}{PQ} = \frac{PQ}{PR}$. Substituting the given values: $\frac{x}{15} = \frac{15}{25}$. Solving for x: $x = 15 \times \frac{15}{25} = 15 \times \frac{3}{5} = 9$ cm..

17 **[Challenging]** • **Target: Transformations of hyperbolas** • **Cross-Lesson: 4.01 Statistics & Probability**

The lifespan X in hours of an electronic component is normally distributed with mean $\mu = 1200$ hours and standard deviation $\sigma = 150$ hours. (a) Find the probability that a randomly selected component lasts more than 1350 hours. (b) In a batch of 500 components, estimate how many will last between 1000 and 1400 hours. (c) The manufacturer guarantees the top 10% longest-lasting components. Determine the minimum lifespan required to receive this guarantee.

(a) Standardize: $Z = \frac{1350 - 1200}{150} = \frac{150}{150} = 1.0$. $P(X > 1350) = P(Z > 1.0) = 1 - \Phi(1.0) \approx 1 - 0.8413 = 0.1587 \quad (15.9\%).$ (b) For $1000 \le X \le 1400$: $Z_1 = \frac{1000 - 1200}{150} = -\frac{200}{150} = -1.3333$. $Z_2 = \frac{1400 - 1200}{150} = \frac{200}{150} = 1.3333$. Using GDC `normalcdf(-1.3333, 1.3333, 0, 1)`: $P \approx 0.8176$. Expected number: $500 \times 0.8176 \approx 408.8 \approx 409$ components. (c) We require.

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18 **[Challenging]** • **Target: Graphing square root functions** • **Cross-Lesson: 5.01 Calculus**

In a factory, 60% of components are manufactured by Machine A and 40% by Machine B. It is known that 3% of components produced by Machine A are defective, whereas 5% of components produced by Machine B are defective. (a) Find the probability that a randomly chosen component is defective. (b) Given that a randomly chosen component is defective, find the probability that it was produced by Machine B. (c) Determine whether the events "component produced by Machine B" and "component is defective" are independent.

(a) By the law of total probability: $P(D) = P(A)P(D \mid A) + P(B)P(D \mid B) = (0.60)(0.03) + (0.40)(0.05)$ $= 0.018 + 0.020 = 0.038 \quad (3.8\%).$ (b) By Bayes' Theorem / conditional probability: $P(B \mid D) = \frac{P(B \cap D)}{P(D)} = \frac{P(B)P(D \mid B)}{P(D)} = \frac{(0.40)(0.05)}{0.038} = \frac{0.020}{0.038} = \frac{20}{38} = \frac{10}{19} \approx 0.5263 \quad (52.6\%).$ (c) Check the independence condition $P(B \cap D) = P(B)P(D)$: $P(B \cap D) = 0.020$. $P(B) \times P(D) = 0.40 \times 0.038 = 0.0152$..

19 **[Challenging]** • **Target: Cubic functions** • **Cross-Lesson: 6.01 AI SL — Specific Topics**

In a 2D Cartesian plane, points A, B, C have coordinates $A(2, -1)$, $B(5, 3)$, and $C(-1, k)$. (a) Find the vectors $\vec{AB}$ and $\vec{AC}$ in component form. (b) Find the value of k for which $\vec{AB}$ is perpendicular to $\vec{AC}$. (c) For $k = 3$, calculate the exact cosine of the angle between $\vec{AB}$ and $\vec{AC}$.

(a) $\vec{AB} = (5 - 2; 3 - (-1)) = (3; 4)$. $\vec{AC} = (-1 - 2; k - (-1)) = (-3; k + 1)$. (b) Perpendicularity condition $\vec{AB} \cdot \vec{AC} = 0$: $3(-3) + 4(k + 1) = 0 \Rightarrow -9 + 4k + 4 = 0 \Rightarrow 4k - 5 = 0 \Rightarrow k = \frac{5}{4} = 1.25.$ (c) When $k = 3$: $\vec{AC} = (-3; 4)$. Lengths: $|\vec{AB}| = \sqrt{3^2 + 4^2} = 5$, $|\vec{AC}| = \sqrt{(-3)^2 + 4^2} = 5$. Dot product: $\vec{AB} \cdot \vec{AC} = 3(-3) + 4(4) = -9 + 16 = 7$..

20 **[Challenging]** • **Target: Power functions** • **Cross-Lesson: 1.01 Number & Algebra**

Consider the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^9$. (a) Write down the general term in terms of r. (b) Find the coefficient of x^3. (c) Find the term independent of x.

(a) General term: $T_{r+1} = \binom{9}{r}(x^2)^{9-r}\left(-\frac{2}{x}\right)^r = \binom{9}{r}(-2)^r x^{2(9-r)}x^{-r} = \binom{9}{r}(-2)^r x^{18-3r}.$ (b) To find the coefficient of x^3, set the power of x equal to 3: $18 - 3r = 3 \Rightarrow 3r = 15 \Rightarrow r = 5.$ Substitute $r = 5$: $T_6 = \binom{9}{5}(-2)^5 x^3 = 126 \times (-32)x^3 = -4032x^3.$ The coefficient of x^3 is -4032. (c) The term independent of x has power 0: $18 - 3r = 0 \Rightarrow 3r = 18 \Rightarrow r = 6.$ Substitute.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Cubic functions

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(2)$.

a. 10 ✓

b. 12

c. 4

d. 14

Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

22 [Foundational] • Target: Cubic functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

23 [Foundational] • Target: Cubic functions

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(5)$.

a. 16 ✓

b. 18

c. 10

d. 20

Substitute $x = 5$: $f(5) = 2(5) + 6 = 16$.

24 [Foundational] • Target: Power functions

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(2)$.

a. 13 ✓

b. 16

c. 6

d. 17

Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

25 [Foundational] • Target: Power functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

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26 [Foundational] • Target: Power functions

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(5)$.

a. 22 ✓

b. 25

c. 15

d. 26

Substitute $x = 5$: $f(5) = 3(5) + 7 = 22$.

27 [Foundational] • Target: Direct variation

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(2)$.

a. 16 ✓

b. 20

c. 8

d. 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

28 [Foundational] • Target: Direct variation

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cubic functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

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30 [Intermediate] • Target: Cubic functions

A data set recorded for cubic functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

31 [Intermediate] • Target: Cubic functions

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(4)$.

a. 14 ✓

b. 16

c. 8

d. 18

Substitute $x = 4$: $f(4) = 2(4) + 6 = 14$.

32 [Intermediate] • Target: Cubic functions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

33 [Intermediate] • Target: Power functions

A data set recorded for power functions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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34 [Intermediate] • Target: Power functions

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(3)$.

a. 16 ✓

b. 19

c. 9

d. 20

Substitute $x = 3$: $f(3) = 3(3) + 7 = 16$.

35 [Intermediate] • Target: Power functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

36 [Intermediate] • Target: Direct variation

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

37 [Intermediate] • Target: Direct variation

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(3)$.

a. 20 ✓

b. 24

c. 12

d. 24

Substitute $x = 3$: $f(3) = 4(3) + 8 = 20$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Direct variation • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(4)$.

a. 24 ✓

b. 28

c. 16

d. 28

Substitute $x = 4$: $f(4) = 4(4) + 8 = 24$.

39 [Challenging] • Target: Direct variation • Cross-Lesson: 6.01 AI SL — Specific Topics

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

40 [Challenging] • Target: Hyperbolas and inverse variation • Cross-Lesson: 1.01 Number & Algebra

A data set recorded for hyperbolas and inverse variation consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

41 [Challenging] • Target: Hyperbolas and inverse variation • Cross-Lesson: 3.01 Geometry & Trigonometry

For the model $f(x) = 5x + 9$ in hyperbolas and inverse variation, evaluate $f(4)$.

a. 29 ✓

b. 34

c. 20

d. 33

Substitute $x = 4$: $f(4) = 5(4) + 9 = 29$.

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42 [Challenging] • Target: Hyperbolas and inverse variation • Cross-Lesson: 4.01 Statistics & Probability

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

43 [Challenging] • Target: Transformations of hyperbolas • Cross-Lesson: 5.01 Calculus

A data set recorded for transformations of hyperbolas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

44 [Challenging] • Target: Transformations of hyperbolas • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 6x + 3$ in transformations of hyperbolas, evaluate $f(4)$.

a. 27 ✓

b. 33

c. 24

d. 31

Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

45 [Challenging] • Target: Graphing square root functions • Cross-Lesson: 1.01 Number & Algebra

For the model $f(x) = 2x + 4$ in graphing square root functions, evaluate $f(3)$.

a. 10 ✓

b. 12

c. 6

d. 14

Substitute $x = 3$: $f(3) = 2(3) + 4 = 10$.

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